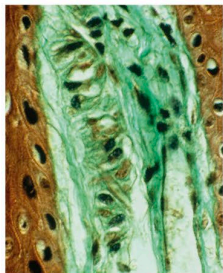


SKIN AND EPITHELIAL TISSUES

SKIN PROTECTS THE UNDERLYING TISSUES AND ALSO PROVIDES THE SENSE OF TOUCH. IT IS A SPECIALIZED TYPE OF EPITHELIUM. EPITHELIAL TISSUES OCCUR THROUGHOUT THE BODY, PROVIDING COVERINGS AND LININGS FOR BODY PARTS AND ORGANS.

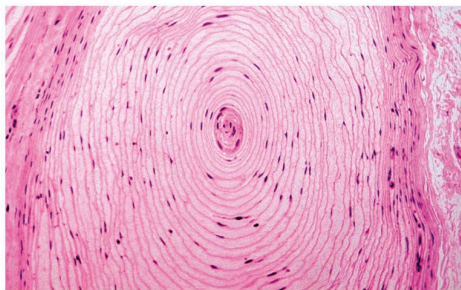
COMPLEXITIES OF TOUCH

The sense of touch is based in the lower of the skin layers, the dermis. Microsensors – the endings of tiny nerve cells – in the dermis detect various physical changes, from light contact to heavy, painful pressure. On average, a skin patch the size of a fingernail contains about 1,000 receptors. However, the skin on the fingertips has more than 3,000 receptors that detect light touch for precise feeling. There are also receptor fibres



LIGHT-TOUCH SENSOR

This microscope view shows a Meissner's corpuscle (green) in a fingertip. It is important for light, discriminatory touch.



DEEP-PRESSURE SENSOR

Pacinian corpuscles have a multi-layered structure and are the largest of the skin receptors, in some areas being more than 1mm (1/25in) long.

around the bases of hairs, in the follicles (pits) within the dermis. Different types of receptor respond more readily to certain types of stimulation, but almost all respond to most stimuli.

Superficial nerve endings

Penetrate the epidermis; occur everywhere in the skin and include free nerve endings

Meissner's corpuscle

Upper dermal nerve ending; mostly located just below the base of the epidermis

Pacinian corpuscle

Located deep in the dermis

